

Systems of Linear Differential Equations

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Condensed Cheat Sheet: Eigenvalue Decoupling, e^{At} , and Fundamental Matrices

Unit 15 at a Glance: Solving Coupled Linear Systems

A coupled system $\mathbf{x}' = A\mathbf{x}$ looks tangled only because the variables are written in the wrong basis. The eigenvectors of A are the right basis: in those coordinates the matrix becomes diagonal, $A = PDP^{-1}$, and the substitution $\mathbf{y} = P^{-1}\mathbf{x}$ *decouples* the system into independent scalar equations $\mathbf{y}' = D\mathbf{y}$. Packaging all of the modes into one operator gives the matrix exponential $e^{At} = Pe^{Dt}P^{-1}$, the exact matrix-valued analogue of the scalar e^{at} , whose columns are collected by the fundamental matrix.

- **Higher order to first order** — any n th-order linear ODE becomes a first-order system $\mathbf{x}' = A\mathbf{x}$ through the state vector of derivatives.
- **Decoupling by diagonalization** — with distinct eigenvalues, $A = PDP^{-1}$ and $\mathbf{y} = P^{-1}\mathbf{x}$ split the system into independent modes $y'_i = \lambda_i y_i$.
- **The matrix exponential** — $e^{At} = Pe^{Dt}P^{-1}$ solves $\mathbf{x}' = A\mathbf{x}$, $\mathbf{x}(0) = \mathbf{x}_0$ as $\mathbf{x}(t) = e^{At}\mathbf{x}_0$, generalizing scalar e^{at} .
- **Fundamental matrices** — $\Psi(t) = [\mathbf{x}_1 \mid \mathbf{x}_2 \mid \cdots]$ collects independent solutions; $\Psi(t)\Psi(0)^{-1} = e^{At}$.
- **Complex eigenvalues** — $\lambda = \alpha \pm i\beta$ produce spiraling real modes $e^{\alpha t}(\cos \beta t, \sin \beta t)$ through Euler's formula.

Part I — Condensed Cheat Sheet

15.1 Converting Higher-Order Equations to First-Order Systems

Recasting an n th-order linear ODE as a first-order system; the state vector of successive derivatives; the companion matrix; the unified matrix form $\mathbf{x}' = A\mathbf{x}$.

The State-Vector Substitution

$$x_1 = y, \quad x_2 = y', \quad \dots, \quad x_n = y^{(n-1)} \Rightarrow x'_1 = x_2, \quad x'_2 = x_3, \quad \dots, \quad x'_{n-1} = x_n.$$

Companion Form of $y'' + py' + qy = 0$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad x_1 = y, \quad x_2 = y'.$$

The Unified Matrix Form

$$\mathbf{x}' = A\mathbf{x}, \quad \mathbf{x} = (x_1, x_2, \dots, x_n)^\top, \quad A \text{ constant } n \times n.$$

Method — Convert an n th-Order ODE to a System

1. Name the state variables $x_1 = y$, $x_2 = y'$, $\dots$, $x_n = y^{(n-1)}$.
2. Differentiate each: the first $n - 1$ rows are simply $x'_i = x_{i+1}$.
3. Solve the original ODE for the top derivative $y^{(n)}$ and write the last row x'_n in terms of the x_i .
4. Collect the coefficients into the companion matrix A and read off $\mathbf{x}' = A\mathbf{x}$.

Key Takeaway

Every higher-order linear ODE is secretly a first-order **system**. The state vector stacks y together with its derivatives, and the companion matrix's characteristic polynomial is **exactly the characteristic equation of the original ODE** — so the eigenvalue machinery and the scalar method agree.

Common Pitfall

When you solve for the top derivative, the coefficients enter the last row of A with **flipped signs**: $y'' = -6y - 5y'$ gives the row $(-6, -5)$, not $(6, 5)$. Forgetting the sign flip silently changes every eigenvalue.

15.2 The Eigenvalue Method and Decoupling

The trial solution $\mathbf{x} = e^{\lambda t}\mathbf{v}$; the modal matrix P of eigenvectors and the diagonal eigenvalue matrix D ; the diagonalization $A = PDP^{-1}$; the modal substitution $\mathbf{y} = P^{-1}\mathbf{x}$ that turns $\mathbf{x}' = A\mathbf{x}$ into the decoupled $\mathbf{y}' = D\mathbf{y}$.

Eigen-Ansatz and the General Solution

$$\mathbf{x} = e^{\lambda t}\mathbf{v} \Rightarrow A\mathbf{v} = \lambda\mathbf{v}; \quad \mathbf{x}(t) = c_1e^{\lambda_1 t}\mathbf{v}_1 + c_2e^{\lambda_2 t}\mathbf{v}_2.$$

Modal Matrix and Diagonalization

$$P = [\mathbf{v}_1 \mid \mathbf{v}_2], \quad D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}; \quad A = PDP^{-1}, \quad D = P^{-1}AP.$$

The Decoupling Substitution

$$\mathbf{y} = P^{-1}\mathbf{x} \Rightarrow \mathbf{y}' = P^{-1}\mathbf{x}' = P^{-1}A\mathbf{x} = P^{-1}AP\mathbf{y} = D\mathbf{y}: \quad y'_i = \lambda_i y_i.$$

Method — Decouple $\mathbf{x}' = A\mathbf{x}$ into $\mathbf{y}' = D\mathbf{y}$

1. Solve $\det(A - \lambda I) = 0$ for the eigenvalues λ_1, λ_2 and find an eigenvector $\mathbf{v}_i$ for each.
2. Build the modal matrix $P = [\mathbf{v}_1 \mid \mathbf{v}_2]$ from the eigenvectors as columns and the diagonal $D = \text{diag}(\lambda_1, \lambda_2)$.
3. Compute P^{-1} and confirm the diagonalization $P^{-1}AP = D$ (this is the algebraic heart of the decoupling).
4. Substitute $\mathbf{y} = P^{-1}\mathbf{x}$: the system collapses to the independent modes $y'_i = \lambda_i y_i$, solve each as $y_i = c_i e^{\lambda_i t}$, then return via $\mathbf{x} = P\mathbf{y}$.

Key Takeaway

Diagonalization *is* decoupling. Writing $A = PDP^{-1}$ and changing to modal coordinates $\mathbf{y} = P^{-1}\mathbf{x}$ replaces the tangled $\mathbf{x}' = A\mathbf{x}$ with n **independent scalar equations** $y'_i = \lambda_i y_i$. Each mode evolves as $e^{\lambda_i t}$ on its own eigenvector, and $\mathbf{x} = P\mathbf{y}$ reassembles them.

Common Pitfall

The eigenvectors are the **columns** of P , in the same order as the eigenvalues down the diagonal of D . Swapping a column without swapping the matching diagonal entry breaks $P^{-1}AP = D$. Diagonalization also **requires a full set of independent eigenvectors** — a repeated eigenvalue short on eigenvectors is defective and needs the generalized-eigenvector route instead.

15.3 The Matrix Exponential

The power-series definition of e^{At} ; its closed form $e^{At} = Pe^{Dt}P^{-1}$ via diagonalization; $e^{Dt} = \text{diag}(e^{\lambda_i t})$; why e^{At} is the natural matrix generalization of the scalar e^{at} ; the initial-value solution $\mathbf{x}(t) = e^{At}\mathbf{x}_0$.

Definition by the Exponential Series

$$e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots, \quad \frac{d}{dt}e^{At} = Ae^{At}.$$

Closed Form via Diagonalization

$$A = PDP^{-1} \Rightarrow e^{At} = Pe^{Dt}P^{-1}, \quad e^{Dt} = \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix}.$$

Generalization of the Scalar e^{at}

$$\text{scalar: } x' = ax \Rightarrow x = e^{at}x_0; \quad \text{matrix: } \mathbf{x}' = A\mathbf{x} \Rightarrow \mathbf{x}(t) = e^{At}\mathbf{x}_0, \quad e^{A \cdot 0} = I.$$

Method — Compute e^{At} by Diagonalization

1. Diagonalize $A = PDP^{-1}$ using the eigenvectors (columns of P) and eigenvalues (diagonal of D).
2. Exponentiate the diagonal entrywise: $e^{Dt} = \text{diag}(e^{\lambda_1 t}, e^{\lambda_2 t})$, since the series of a diagonal matrix acts independently on each entry.
3. Reassemble $e^{At} = P e^{Dt} P^{-1}$ by multiplying the three matrices left to right.
4. Sanity check $e^{A \cdot 0} = I$, then read off the IVP solution $\mathbf{x}(t) = e^{At} \mathbf{x}_0$.

Key Takeaway

The matrix exponential is the **single operator that advances the whole system in time**. Because A and P share eigenvectors, $e^{At} = P e^{Dt} P^{-1}$ simply exponentiates each eigenvalue and reassembles — the exact matrix analogue of e^{at} , satisfying $\frac{d}{dt} e^{At} = A e^{At}$ and $e^{A \cdot 0} = I$, so $\mathbf{x}(t) = e^{At} \mathbf{x}_0$ solves the IVP outright.

Common Pitfall

e^{At} is **not** the entrywise exponential of At — you must use the series or the diagonalization. And $e^{At} e^{Bt} = e^{(A+B)t}$ holds **only when A and B commute** ($AB = BA$); for non-commuting matrices that shortcut fails.

15.4 Fundamental Matrices and the General Solution

The fundamental matrix $\Psi(t)$ whose columns are independent solutions; the Wronskian as its determinant; the compact general solution $\mathbf{x} = \Psi(t)\mathbf{c}$; the link $e^{At} = \Psi(t)\Psi(0)^{-1}$ that normalizes any fundamental matrix to the exponential.

Fundamental Matrix and General Solution

$$\Psi(t) = [\mathbf{x}_1(t) \mid \mathbf{x}_2(t)], \quad \Psi'(t) = A \Psi(t), \quad \mathbf{x}(t) = \Psi(t) \mathbf{c}.$$

Independence via the Wronskian

$W(t) = \det \Psi(t) \neq 0$ for some $t \Rightarrow$ the columns are linearly independent solutions.

Normalization to the Matrix Exponential

$$e^{At} = \Psi(t) \Psi(0)^{-1}, \quad e^{At}|_{t=0} = \Psi(0) \Psi(0)^{-1} = I.$$

Method — Build and Use a Fundamental Matrix

1. Find a full set of independent solutions $\mathbf{x}_i(t) = e^{\lambda_i t} \mathbf{v}_i$ from the eigenpairs.
2. Place them as the columns of $\Psi(t) = [\mathbf{x}_1 \mid \mathbf{x}_2 \mid \cdots]$ and confirm $\det \Psi \neq 0$.
3. Write the general solution compactly as $\mathbf{x}(t) = \Psi(t)\mathbf{c}$, fixing $\mathbf{c}$ from $\mathbf{x}(0) = \Psi(0)\mathbf{c}$.
4. If the transition operator is wanted, normalize: $e^{At} = \Psi(t)\Psi(0)^{-1}$.

Key Takeaway

The fundamental matrix is the **bookkeeping device for an entire solution space**: its columns are independent solutions, $\det \Psi \neq 0$ guarantees they span, and $\mathbf{x} = \Psi\mathbf{c}$ is every solution at once. Normalizing it at $t = 0$ recovers the matrix exponential, $e^{At} = \Psi(t)\Psi(0)^{-1}$ — two languages for the same flow.

Common Pitfall

Any Ψ with independent columns is a fundamental matrix, but it equals e^{At} **only when** $\Psi(0) = I$. If $\Psi(0) \neq I$ you must multiply by $\Psi(0)^{-1}$ on the right — not the left. Order matters: $\Psi(t)\Psi(0)^{-1}$, never $\Psi(0)^{-1}\Psi(t)$.

15.5 Complex Eigenvalues and Oscillatory Modes

Complex conjugate eigenvalues $\lambda = \alpha \pm i\beta$; extracting two real solutions from one complex eigensolution via Euler's formula; the real general solution as decaying or growing oscillations $e^{\alpha t}(\cos \beta t, \sin \beta t)$.

Complex Eigenpair and Euler Expansion

$$\lambda = \alpha \pm i\beta : \quad e^{\lambda t} \mathbf{v} = e^{\alpha t} (\cos \beta t + i \sin \beta t) \mathbf{v}.$$

Two Real Solutions from One Complex Mode

$$\mathbf{x}_1 = \operatorname{Re}(e^{\lambda t} \mathbf{v}), \quad \mathbf{x}_2 = \operatorname{Im}(e^{\lambda t} \mathbf{v}); \quad \mathbf{x} = c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2.$$

Geometry from the Sign of α

$$\alpha < 0 : \text{inward spiral}, \quad \alpha = 0 : \text{closed orbits (center)}, \quad \alpha > 0 : \text{outward spiral}.$$

Method — Solve a System with Complex Eigenvalues

1. Find the complex eigenvalue $\lambda = \alpha + i\beta$ and a complex eigenvector $\mathbf{v}$ (use only one of the conjugate pair).
2. Form the complex solution $e^{\lambda t}\mathbf{v}$ and expand with $e^{i\beta t} = \cos \beta t + i \sin \beta t$.
3. Split into real and imaginary parts $\mathbf{x}_1 = \text{Re}$, $\mathbf{x}_2 = \text{Im}$ — two independent real solutions.
4. Write the real general solution $\mathbf{x} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2$ and read the spiral direction from $\text{sign}(\alpha)$.

Key Takeaway

A complex eigenpair encodes **rotation plus scaling**. Euler's formula turns the single complex mode $e^{(\alpha+i\beta)t}\mathbf{v}$ into two real solutions whose $\cos \beta t$ and $\sin \beta t$ supply the oscillation and whose $e^{\alpha t}$ envelope sets the spiral: inward for $\alpha < 0$, a closed center for $\alpha = 0$, outward for $\alpha > 0$.

Common Pitfall

Use **only one** eigenvalue of the conjugate pair — taking both double-counts the solution space. And the real and imaginary parts are *already* the two independent real solutions; do not also tack on the conjugate $\bar{\lambda}$ mode.